

Route G2-R: Joint Output Modes and Detector Recoil

Conditional tree-level predictions from the unchanged G2 action

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Abstract

We turn the existing G2 collision amplitudes into joint mode-recoil predictions, keeping all open channels and distinguishing scattering events from mode-conversion events. A mass-only detector cannot resolve the sign of its compact momentum. For nonelastic channels in the specified sharp-beam preparation, outgoing ordinary momentum can disambiguate that sign. This is a model-dependent reconstruction, not an independent proof of compact-momentum conservation. No new action, parameter fit, absolute encounter probability or external drive is added.

1 Inputs and two different event ensembles

The G2 interaction remains the positive local potential $\kappa_5|\Phi|^2|X|^2$ on $\mathbb{M}^{3,1} \times S^1$. Its normalized circle overlap gives

$$m_n^2 = M_5^2 + (n + \alpha)^2/R^2, \quad \mu_l^2 = M_D^2 + l^2/R^2, \quad \mathcal{M} = -g \delta_{n+l, m+j}, \quad g = \kappa_5/(2\pi R). \quad (1)$$

Keep the entire G2 engineering card: $R = 1$, $M_5 = 0.5$, $\alpha = 0.25$, $M_D = 5$, $g = 0.05$, incoming Φ label $n = 1$, and ordinary center-of-mass momentum $p_i = 0.2$. The detector coefficients are exactly finite,

$$c_l = \mathcal{N} e^{-l^2/(4w^2)} e^{-il\theta_0}, \quad l = -4, \dots, 4, \quad w = 1.2, \quad \theta_0 = 0.7, \quad w_l = |c_l|^2, \quad \sum_l w_l = 1. \quad (2)$$

These are prepared-state inputs, not empirical fitted values. For each incoming l , set

$$\sqrt{s_l} = \sqrt{m_n^2 + p_i^2} + \sqrt{\mu_l^2 + p_i^2}, \quad j = n + l - m, \quad (3)$$

$$p_{lm} = \frac{\sqrt{[s_l - (m_m + \mu_j)^2][s_l - (m_m - \mu_j)^2]}}{2\sqrt{s_l}}, \quad (4)$$

$$K_{lm} = \sigma_{lm} v_{M,l} = \frac{g^2 p_{lm}}{16\pi E_n E_l \sqrt{s_l}}. \quad (5)$$

Here $K_{lm} = 0$ unless $\sqrt{s_l} > m_m + \mu_j$; closed channels must be rejected before evaluating the radical. The distinct-particle phase-space convention is the same as G2 and the PDG review [1]. Define

$$\mathcal{A} = \{(l, m) : w_l > 0, \sqrt{s_l} > m_m + \mu_{n+l-m}\}, \quad \mathcal{C} = \{(l, m) \in \mathcal{A} : m \neq n\}. \quad (6)$$

$\mathcal{A}$ contains all *scattering events*, including elastic $m = n$ collisions. $\mathcal{C}$ contains only mode-changing events. Neither includes the unscattered identity amplitude as a collision. This distinction is required before assigning a normalized fraction.

2 Joint distribution, not a single target fraction

For $B = \mathcal{A}$ or $\mathcal{C}$, let

$$Z_B = \sum_{(l,m) \in B} w_l K_{lm}, \quad \boxed{P_B(m, j) = \frac{w_{m+j-n} K_{m+j-n, m}}{Z_B} \mathbf{1}_{\{(m+j-n, m) \in B\}}}. \quad (7)$$

For $Z_B > 0$ these dimensionless quantities sum to one and are nonnegative. They are probabilities *conditional on the selected type of scattering event*, in the common-overlap dilute approximation of G2. They are not the probability that an arbitrary incident packet scatters. For unequal encounter overlaps I_l , replace $w_l K_{lm}$ by $w_l I_l K_{lm}$ and renormalize. No values of I_l are invented here. The common coupling g^2 cancels from (7), but not from the event rate.

The marginal and conditional predictions are

$$P_B^\Phi(m) = \sum_j P_B(m, j), \quad P_B^X(j) = \sum_m P_B(m, j), \quad P_B(j | m) = \frac{P_B(m, j)}{P_B^\Phi(m)} \quad (8)$$

where the denominator is nonzero. The complete table, rather than only $m = 0$, is saved in the numerical report. Conditional on each incoming l , support lies on $m + j = n + l$. A packet with several l therefore need not exhibit a single perfectly anticorrelated line.

Event conditioning changes the incoming detector distribution to

$$P_B^{\text{in}}(l) = \frac{w_l \sum_{m: (l, m) \in B} K_{lm}}{Z_B}. \quad (9)$$

Conservation identities must use this distribution:

$$\langle m \rangle_B + \langle j \rangle_B - \langle l \rangle_B = n, \quad (10)$$

$$\text{Var}_B(l) = \text{Var}_B(m) + \text{Var}_B(j) + 2 \text{Cov}_B(m, j), \quad (11)$$

$$\langle j - l | m \rangle_B = n - m. \quad (12)$$

Comparing selected outgoing recoil with the *unconditioned* incident mean would mix momentum transfer with event-selection bias. Perfect anticorrelation applies to the transfers $\Delta j = j - l = -\Delta m = n - m$, so $\text{Cov}(\Delta m, \Delta j) = -\text{Var}(\Delta m)$. It does not generally apply to the output labels m, j themselves.

For the declared card there are 25 open (l, m) channels. The total and mode-changing rate coefficients are respectively $Z_{\mathcal{A}} = 1.85661445 \cdot 10^{-6}$ and $Z_{\mathcal{C}} = 1.63742848 \cdot 10^{-6}$. The all-scattering output marginal is

Output m	-2	-1	0	1 (elastic)
$P_{\mathcal{A}}^\Phi(m)$	0.00785985	0.27119405	0.60288929	0.11805680

Thus $Z_{\mathcal{C}}/Z_{\mathcal{A}} = 0.88194320$ of scattering events change mode under this prescription. This is not an incident conversion efficiency.

For fixed (m, j) , $l = m + j - n$ is fixed, and the ideal narrow-beam momentum-angle law is

$$dP_B(m, j, p, \Omega) = P_B(m, j) \delta(p - p_{lm}) dp \frac{d\Omega}{4\pi}. \quad (13)$$

The two ordinary outgoing momenta are $p_{lm} \hat{\mathbf{u}}$ and $-p_{lm} \hat{\mathbf{u}}$ in the common COM frame. Isotropy is a tree contact-interaction prediction. Finite beam widths and instrumental response would broaden these ideal lines.

3 What a mass-only detector actually resolves

The detector mass $\mu_j = \sqrt{M_D^2 + j^2/R^2}$ measures $r = |j|$, not its sign. The corresponding joint distribution is

$$\bar{P}_B(m, r) = \begin{cases} P_B(m, 0), & r = 0, \\ P_B(m, r) + P_B(m, -r), & r > 0. \end{cases} \quad (14)$$

This is an incoherent sum over orthogonal recoil states. Degeneracy of their measured masses does not make their amplitudes add. Where both signs have support,

$$P_B(j \mid m, r) = \frac{P_B(m, j)}{\bar{P}_B(m, r)}, \quad |j| = r. \quad (15)$$

For $\alpha = 0.25$ the Φ masses determine their signed mode labels uniquely in the ideal tree spectrum: an equality for distinct m, m' would require $m + m' = -2\alpha$, impossible for integers. At $2\alpha \in \mathbb{Z}$ this step can fail because of Φ degeneracies. G1 already requires eigenspace, not arbitrary basis-state, interpretations at such degeneracies.

Only diagonal recoil event rates are calculated. They are independent of θ_0 , and an incoming detector mixture diagonal in l with the same w_l gives the same rates. Thus this joint table is not an entanglement witness or evidence that the preparation was localized. It also does not construct a real measurement apparatus for signed j .

4 A kinematic way to resolve the sign in conversion events

Suppose the ideal readout provides (m, r, p_f) , the parameters and incoming p_i are known, and the event is in the COM frame. The measured final energy and inferred incoming detector momentum are

$$E_f = \sqrt{m_m^2 + p_f^2} + \sqrt{M_D^2 + r^2/R^2 + p_f^2}, \quad (16)$$

$$E_n = \sqrt{m_n^2 + p_i^2}, \quad (17)$$

$$l^2 = R^2 [(E_f - E_n)^2 - M_D^2 - p_i^2]. \quad (18)$$

Write $d = m - n$ so that $l = d + j$. If $d \neq 0$,

$$l^2 = d^2 + 2dj + r^2 \implies \boxed{j = \frac{l^2 - d^2 - r^2}{2d}}. \quad (19)$$

The result must be consistent with $j = \pm r$, integer mode labels, the prepared support and the open-channel condition. This is *model-dependent kinematic reconstruction*: the dispersion law and momentum conservation were used to obtain it. Checking that the reconstructed j obeys the same conservation law is not an independent test of that law. The independently comparable predictions are the joint recoil line positions, event fractions and angular distribution.

The uniqueness can also be proved without inversion. For the two candidate signs $j = \pm r$, the initial labels are $l = d \pm r$, and

$$\sqrt{s_+} - \sqrt{s_-} = \frac{4dr/R^2}{\sqrt{M_D^2 + p_i^2 + (d+r)^2/R^2} + \sqrt{M_D^2 + p_i^2 + (d-r)^2/R^2}}. \quad (20)$$

This is nonzero whenever $dr \neq 0$. At fixed final masses,

$$\frac{dE_f}{dp_f} = p_f \left(\frac{1}{E'_\Phi} + \frac{1}{E'_X} \right) > 0 \quad (21)$$

above threshold, so both open signs have different ordinary outgoing momenta. For elastic $d = 0$, $l = j$ and $p_f = p_i$; these measurements cannot recover the sign. For $r = 0$, the only possible value is already $j = 0$.

A concrete two-line prediction

For the selected output $m = 0$, detector mass $\mu_1 = 5.09901951$ corresponds to $r = 1$. Two allowed components give

Incoming l	Outgoing j	$\sqrt{s_l}$	p_f	$P(j \mid m = 0, r = 1)$
0	+1	6.36506416	1.02175458	0.77457541
-2	-1	6.74994319	1.36191317	0.22542459

These conditional weights use the same prepared packet and common overlap, with equal ideal acceptance. The mass-only readout merges the lines; the ideal momentum readout separates them by $\Delta p_f = 0.34015860$ in the common energy unit. For a deterministic bound $|p_{\text{meas}} - p_{\text{true}}| \leq \epsilon_p$, with all other inputs treated as exact, a sufficient disjoint-interval criterion is $2\epsilon_p < \Delta p_f$. This is a required resolution, not an assumed experimental error bar. Uncertain incoming momentum, parameters, mass assignment or frame must also be propagated before using that criterion experimentally.

5 Readout boundaries and a bounded next step

A calibrated classical readout response $\mathcal{R}(b \mid m, j, p, \Omega)$ would give

$$P_B^{\text{obs}}(b) = \sum_{m,j} P_B(m, j) \int \frac{d\Omega}{4\pi} \mathcal{R}(b \mid m, j, p_{m+j-n,m}, \Omega). \quad (22)$$

Include a missed-event outcome to make $\sum_b \mathcal{R} = 1$. Conditioning on detected events requires renormalization and may change fractions. Only the ideal signed-label and mass-only readouts are evaluated here; no detector efficiency, angular cut or noise distribution is fitted or invented.

The executable imports the unchanged G2 card, enumerates every open channel, and verifies normalization, marginals, Bayes identities, event-conditioned recoil, angular normalization, gauge relabeling, sign folding and reconstruction. Reports preserve the code/input hashes and the complete joint matrices. The underlying G2 tree-EFT and common-overlap limitations remain. In particular no interacting loop error or absolute four-dimensional packet encounter probability is supplied.

The next useful choice is a specified finite-resolution readout to test whether these lines survive realistic input spread and response. If an absolute encounter probability is needed instead, specify normalized four-dimensional packets and use the existing G2 scattering amplitude. An energy-accounted driven background remains an unused backup. Neither option demands a new mass fit, lattice scan or action.

References

- [1] Particle Data Group, *Kinematics*, Review of Particle Physics (2024), sections 49.5–49.5.1, official review.